

```
pip install nltk scikit-learn
```

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Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
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Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.
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Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (f
```

```
import nltk
import string
import numpy as np
```

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

```
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
True
```

```
faq_data = {
    "What is Python?":
        "Python is a programming language used for web development, AI, machine learning, and data s

    "What is NLP?":
        "NLP stands for Natural Language Processing. It helps computers understand human language.",

    "What is Machine Learning?":
        "Machine Learning is a branch of Artificial Intelligence where machines learn from data.",

    "What is Google Colab?":
        "Google Colab is a cloud-based platform used to run Python notebooks online.",

    "What is cosine similarity?":
        "Cosine similarity measures how similar two text sentences are.",

    "What is AI?":
        "AI stands for Artificial Intelligence. It enables machines to mimic human intelligence.",

    "What is Data Science?":
        "Data Science is the process of extracting insights from data using statistics and machine l
}
```

```
def preprocess(text):

    text = text.lower()

    text = text.translate(str.maketrans('', '', string.punctuation))
```

```

words = word_tokenize(text)

stop_words = set(stopwords.words('english'))

words = [word for word in words if word not in stop_words]

return " ".join(words)

```

```

questions = list(faq_data.keys())

processed_questions = [preprocess(q) for q in questions]

print(processed_questions)

['python', 'nlp', 'machine learning', 'google colab', 'cosine similarity', 'ai', 'data science']

```

```

vectorizer = TfidfVectorizer()

question_vectors = vectorizer.fit_transform(processed_questions)

```

```

def chatbot(user_input):

    processed_input = preprocess(user_input)

    input_vector = vectorizer.transform([processed_input])

    similarity = cosine_similarity(input_vector, question_vectors)

    best_match = np.argmax(similarity)

    score = similarity[0][best_match]

    if score > 0.2:
        return faq_data[questions[best_match]]
    else:
        return "Sorry, I could not understand your question."

```

```

print("==== FAQ CHATBOT ====")
print("Type 'exit' to stop chatbot\n")

while True:

    user_question = input("You: ")

    if user_question.lower() == 'exit':
        print("Chatbot: Goodbye!")
        break

    response = chatbot(user_question)

    print("Chatbot:", response)

```

```

==== FAQ CHATBOT ====
Type 'exit' to stop chatbot

```

```

You: what is python
Chatbot: Python is a programming language used for web development, AI, machine learning, and dat
You: what is ai
Chatbot: AI stands for Artificial Intelligence. It enables machines to mimic human intelligence.
You: eit

```

Chatbot: Sorry, I could not understand your question.
You: exit
Chatbot: Goodbye!

Start coding or generate with AI.